

hw 0525

15-4

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a.  $H_0$ : 兩物體之間沒有顯著差異  
 $H_{a1}$ : 有

$$T^+ = 1.5 + 7 + 4 + 5.5 + 3 + 5.5$$

$$T^- = 1.5$$

$$T = \min(T^+, T^-) = 1.5$$

critical value ( $n=7, \alpha=0.05$  雙尾) = 2

$$T = 1.5 \leq 2 \Rightarrow \text{Reject } H_0$$

b.  $p\text{-value} = P(X \leq 1) + P(X \geq 6)$

$$= \binom{7}{0} \left(\frac{1}{2}\right)^7 + \binom{7}{6} \left(\frac{1}{2}\right)^7 = 0.025$$

$\therefore p\text{-value} > 0.05 \Rightarrow \text{Non-Reject } H_0$ , 不同於 (a)

|      |     |     |      |     |     |     |
|------|-----|-----|------|-----|-----|-----|
| 1.5  | 7   | 4   | 1.5  | 5.5 | 3   | 5.5 |
| 1    | 2   | 3   | 4    | 5   | 6   | 7   |
| 1    | 2   | 3   | 4    | 5   | 6   | 7   |
| -0.1 | 0.9 | 0.3 | -0.1 | 0.5 | 0.2 | 0.5 |

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Month A B

|   |   |     |
|---|---|-----|
| 1 | 4 | 6   |
| 2 | 2 | 3.5 |
| 3 | 5 | 3.5 |
| 4 | 3 | 7   |
| 5 | 0 | 5   |
| 6 | 1 | 1.5 |
| 7 | 1 | 1.5 |
| 8 | 6 | 8   |
| 9 |   |     |

a.  $H_0$ : 機器 A, B 的月故障率沒有顯著差異  
 $H_{a1}$ : 有

$$T^+ = 1.5 + 1.5 + 3.5 = 6.5$$

$$T^- = 3.5 + 5 + 6 + 7 + 8 = 29.5$$

$$T = \min(6.5, 29.5) = 6.5$$

Critical Value ( $n=8, \alpha=0.05$  雙尾) = 4

$$T = 6.5 > 4 \Rightarrow \text{Non-Reject } H_0$$

b. 產量與工作負荷的月變化、  
 季節及環境因素、原料差異、  
 是否周小保養完

15)

A. Paired t-test, Sign test, Wilcoxon Signed-rank test

- ① 配對差值須來自一呈常態分佈的母體
- ② 樣本數過少, 不能完美符合常態分佈

b.  $H_0$ : 兩母體沒有顯著差異

$H_a$ : " 有 "

Sign test:  $+: > 0, -: < 0, n = 20$

$$X = 0$$

$$p\text{-value} = 2 \times P(X \leq 0) = 2 \times 0.5^{20} < 0.05$$

$\Rightarrow$  reject  $H_0$

Wilcoxon Signed-rank test:

$$T^+ = 10, T^- = 0 \Rightarrow T = 0$$

$$\text{Critical Value} = 52$$

$$T = 0 < 52 \Rightarrow \text{reject } H_0$$

c. (b) 兩個結論相同

15.5

(b)

Rank:

|    | 5  | 1.0 | 1.5 | 2.0 |
|----|----|-----|-----|-----|
| 13 | 16 | 2   | 2   |     |
| 20 | 10 | 2.5 | 10  |     |
| 16 | 5  | 10  | 5   |     |
| 19 | 13 | 13  | 5   |     |
| 18 | 16 | 2.5 | 2   |     |

Rank 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20

$H_0$ : 四種不同訓練時間組別的積木組裝時間分佈沒有差異

$H_a$ : 至少有一組分佈和其他組不同

$$H = \frac{12}{N(N+1)} \sum \frac{R_i^2}{n_i} - 3(N+1)$$

$$= \frac{12}{20(21)} \left[ \frac{1396}{5} + \frac{3600}{5} + \frac{1600}{5} + \frac{596}{5} \right] - 3(21)$$

$$= 12.27$$

$$df = k - 1 = 4 - 1 = 3, \alpha = 0.01, \chi_{3,0.01}^2 = 11.345$$

$$12.27 > 11.345, \text{Reject } H_0$$

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- a.  $H_0$ : 四組母體分佈相同  
 $H_a$ : 不完全相同

$$H = \frac{12}{N(N+1)} \sum \frac{T_i^2}{n_i} - 3(N+1)$$

$$= \frac{12}{40 \times 41} \left[ \frac{247.5^2}{10} + \frac{168^2}{10} + \frac{216.5^2}{10} + \frac{188^2}{10} \right]$$

$$= 2.632$$

$$df = 4 - 1 = 3$$

$$\chi^2_{3,0.01} = 11.345$$

$$H < \chi^2_{3,0.01} \Rightarrow \text{Non-reject } H_0$$

b.  $\chi^2_{3,0.1} = 6.251$

$$\Rightarrow p\text{-value} > 0.1 \Rightarrow p\text{-value} \approx 0.452$$

c. ANOVA:  $p\text{-value} = 0.468$   
 H-test:  $p\text{-value} = 0.452$  )  $\Rightarrow$   $\text{Reject } H_0$

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a. Randomized Block Design

Rank

- b. 常態性假設大致滿足  
 變異數同質性不符合

|    |    |    |    |
|----|----|----|----|
| 2  | 1  | 4  | 3  |
| 2  | 3  | 1  | 4  |
| 3  | 1  | 2  | 4  |
| 2  | 4  | 1  | 3  |
| 3  | 4  | 2  | 1  |
| 12 | 13 | 14 | 15 |

c.  $H_0$ : 四種母體分佈相同

$H_a$ : 母體分佈不完全相同

$$Fr = \frac{12}{5 \times 4 \times 5} [12^2 + 13^2 + 14^2 + 15^2] - 3 \times 5 \times 5$$

$$= 1.56$$

$$1.56 < \chi^2_{3,0.05} = 7.815$$

$$\Rightarrow \text{Non-reject } H_0$$

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|   | Rank |   |
|---|------|---|
| 1 | 2    | 3 |
| 4 | 3    | 3 |
| 5 | 4    | 2 |
| 3 | 5    | 5 |
| 2 | 1    | 1 |

$R_A = 18$   
 $R_B = 12$   
 $R_C = 26$   
 $R_D = 27$   
 $R_E = 7$

a.  $H_0$ : 五種不同小麥的機率分佈完全相同  
 $H_a$ : 不完全相同

$$Fr = \frac{12}{b k(k+1)} \sum R_i^2 - 3b(k+1)$$

$$= \frac{12}{6 \times 5 \times (5+1)} (18^2 + 12^2 + 26^2 + 27^2 + 7^2) - 3 \times 6 \times (5+1)$$

$$= 21.133 > \chi^2_{4, 0.05} = 9.488$$

$\Rightarrow$  Reject  $H_0$

$\Rightarrow$  五種小麥品種的產量有顯著差異

b.  $p\text{-value} = 0.000 < 0.05 \Rightarrow$  Reject  $H_0$   
 紅輪同(a)

hw 0601

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⑪

| Water | Rank |
|-------|------|
| 1     | 1.5  |
| 2     | 4    |
| 3     | 3    |
| 4     | 12   |
| 5     | 10   |
| 6     | 9    |
| 7     | 8    |
| 8     | 7    |
| 9     | 6    |
| 10    | 5    |
| 11    | 4    |
| 12    | 3    |

a. f

b. t